Experiment 5 — Pointer Manipulation

Go Program

```
package main

import "fmt"

// Function that accepts a pointer and modifies the original variable
func modifyValue(x *int) {
    *x = *x + 10
}

type Student struct {
    Name string
    Age  int
}

func main() {

    // (1) Pointer referencing and dereferencing
    num := 20

    fmt.Println("Original value:", num)
    fmt.Println("Address of num:", &num)
    fmt.Println("Value using pointer:", *(&num))

    // (2) Passing pointer to a function
    fmt.Println("\nBefore function call:", num)

    modifyValue(&num)

    fmt.Println("After function call:", num)

    // (3) Allocating a struct using new()
    student := new(Student)

    // Accessing and modifying fields through pointer
    student.Name = "Rahul"
    student.Age = 20

    fmt.Println("\nStudent details:")
    fmt.Println("Name:", student.Name)
    fmt.Println("Age:", student.Age)

    // Modifying struct field through pointer
    student.Age = 21

    fmt.Println("Updated Age:", student.Age)
}
```

Expected Output

```
Original value: 20
Address of num: 0xc0000...
Value using pointer: 20

Before function call: 20
After function call: 30

Student details:
Name: Rahul
Age: 20
Updated Age: 21
```